PHY 121 Short & Conceptual Questions Solutions

Complete Textbook-Cited Answers for All Part (a) & Short Questions (≤ 10–15 Marks)
Grouped by Field: (A) Optics · (B) Thermal Physics · (C) Waves & Oscillations
Based on *Physics for Engineers* by Dr. Giasuddin Ahmad & BUET Lecture Notes (2009–2025)

Comprehensive BUET L-1/T-1 Exam Reference

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Part I

A) Optics: Short & Conceptual Questions

1 Defects of Images (Lens Aberrations)

1.1 Aberration Definition, Causes, Classification & Spherical Aberration

Exam Appearances: `2011-12 Q4a (15), 2012-13 Q6c (10), 2014-15 Q7a (14), 2015-16 Q4a (8), 2017-18 Q4a (10), 2018-19 Q4a (10), 2020-21 Q4a (10), 2022-23 Q7a (8), 2024-25 Q1a,b (15).`
Textbook Citations: *Physics for Engineers – Part 1*, Dr. Giasuddin Ahmad, Ch. "Defects of Images / Aberrations of Lenses"; Dr. Md. Azizar Rahman, BUET PHY 121 Lecture Slides (*All Lecture of Defect of images.pdf*, Slides 25–34); N. Subrahmanyam, Brij Lal & M. N. Avadhanulu, *A Textbook of Optics*, Ch. 9.

1. What is meant by the term "Aberration" (Defect of Image)? What are the common defects in the images produced by a single lens?

- **Definition:** The deviations of the actual image formed by an optical system (such as a single lens) from the ideal paraxial predictions in terms of size, shape, sharpness, and position are called **defects of images** or **aberrations**.
- **Causes of Aberrations:**
  1. The Gaussian lens formula $\frac{1}{v} - \frac{1}{u} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ is derived assuming that the aperture of the lens is very small and all incident rays are **paraxial** ($\sin\theta \approx \tan\theta \approx \theta$). In practice, lenses have wide apertures and admit both paraxial and marginal rays.
  2. Real objects are extended rather than point objects lying strictly on the principal axis.
  3. Because the refractive index μ of glass varies with wavelength λ (dispersion), the focal length f of a lens varies with the color of light.
- **Classification of Common Lens Defects:**
  1. **Monochromatic Aberrations (Seidel Aberrations):** Occur even with monochromatic (single-wavelength) light due to lens geometry and wide aperture:
     (a) *Spherical Aberration*
     (b) *Coma (Comatic Aberration)*
     (c) *Astigmatism*
     (d) *Curvature of Field*
     (e) *Distortion (Pin-cushion and Barrel-shaped)*
  2. **Chromatic Aberrations:** Occur only when polychromatic (white) light is used, due to material dispersion ($\mu_v > \mu_r$):
     (a) *Longitudinal (Axial) Chromatic Aberration*
     (b) *Lateral (Transverse) Chromatic Aberration*

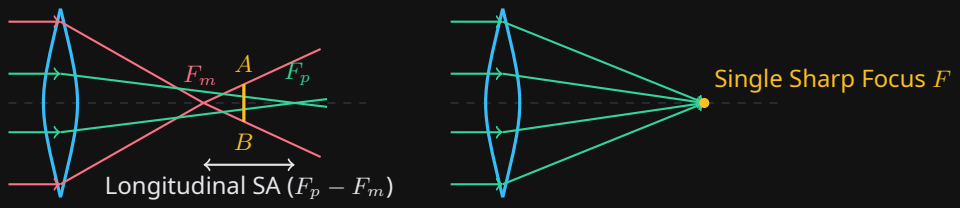
2. Define Spherical Aberration, Longitudinal Spherical Aberration, and Circle of Least Spherical Aberration.

- **Spherical Aberration:** The failure of a lens with spherical surfaces to bring parallel (or point-source) monochromatic rays passing through different zones of the lens to a single focus is called **spherical aberration**. Because a lens acts like a stack of truncated prisms of increasing angle toward the periphery, **marginal rays** (near the outer rim) suffer greater

deviation and focus closer to the lens at I_m (or F_m), whereas **paraxial rays** (close to the principal axis) suffer less deviation and focus farther away at I_p (or F_p).

- **Longitudinal Spherical Aberration:** The axial distance $x = I_p - I_m$ (or $F_p - F_m$) measured along the principal axis between the focus of paraxial rays (I_p) and the focus of marginal rays (I_m).
- **Circle of Least Spherical Aberration (Circle of Least Confusion):** If a screen is placed perpendicular to the optic axis between I_m and I_p, the illuminated patch is circular everywhere and has a minimum diameter AB at a plane XY where the outer marginal cone intersects the inner paraxial cone. This minimum circular cross-section is called the **circle of least spherical aberration** (or circle of least confusion), and its diameter measures the **lateral (transverse) spherical aberration**.

(a) Spherical Aberration in a Real Lens **(b) Image by an Ideal Lens**



3. How can Spherical Aberration be Reduced / Minimized?

1. **Using Stops (Apertures):** Placing an opaque circular diaphragm (stop) in front of the lens blocks either the marginal rays (admitting only paraxial rays) or the paraxial rays, bringing the remaining rays to a sharp focus (though image brightness is reduced).
2. **Using a Crossed Lens (Bending of the Lens):** Spherical aberration is minimum when the total deviation of a ray is divided equally between the two surfaces of the lens. The ratio of the radii of curvature for minimum spherical aberration is:

$$\frac{R_1}{R_2} = \frac{\mu(2\mu - 1) - 4}{\mu(2\mu + 1)}$$

   For glass of $\mu = 1.5$, $\frac{R_1}{R_2} = -\frac{1}{6}$, which is called a **crossed lens**.
3. **Using a Plano-Convex Lens:** In optical instruments (eyepieces and objectives), a plano-convex lens is placed with its **curved (convex) surface facing the incident (or emergent) parallel rays**, which divides the deviation nearly equally between the two surfaces. (If the plane face faces the parallel beam, zero deviation occurs at the first face and all deviation occurs at the second face, maximizing aberration).
4. **Using Two Converging Lenses Separated by a Finite Distance d:** Two thin plano-convex lenses of focal lengths f_1 and f_2 separated by a distance equal to the difference of their focal lengths ($\boxed{d = f_1 - f_2}$) divide the deviation equally ($\delta_1 = \delta_2$) and minimize spherical aberration.

1.2 Coma & Curvature of Field (Petzval's Condition)

Exam Appearances: `2014-15 Q7a (14)`, `2021-22 Q6a (10)`.
Textbook Citations: *Physics for Engineers – Part 1*, Dr. Giasuddin Ahmad, Ch. "Defects of Images"; Dr. Md. Azizar Rahman, BUET PHY 121 Lecture Slides (*All Lecture of Defect of images.pdf*, Slides 37–39, 46–47).

1. What is Coma (Comatic Aberration)? How can it be minimized?

- **Definition of Coma:** While spherical aberration applies to object points on the principal axis, **coma** is the monochromatic aberration affecting rays from an **off-axis point object**. Because the linear magnification varies for rays passing through different concentric circular zones of the lens, each zone forms a circular ring image of progressively larger radius and shifted center, producing a comet-like flare (bright nucleus with a tail).
- **Minimization of Coma:**
  1. **Abbe's Sine Condition:** Coma is completely eliminated if the lateral magnification $\frac{y_2}{y_1}$ is constant for all zones of the lens, which requires satisfying **Abbe's Sine Condition**:

$$\boxed{\mu_1 y_1 \sin\theta_1 = \mu_2 y_2 \sin\theta_2 = \text{constant} \implies \frac{\sin\theta_1}{\sin\theta_2} = \text{constant}}$$

  where μ_1, y_1, θ_1 and μ_2, y_2, θ_2 are the refractive indices, heights, and ray slope angles in the object and image spaces, respectively.
  2. **Use of Aplanatic Lenses & Stops:** A lens designed with conjugate points free from both spherical aberration and coma is called an **aplanatic lens**; coma can also be minimized by placing a suitable stop at the proper position along the axis.

2. Explain the Defect "Curvature of the Field" and How It Can Be Minimized (`2021-22 Q6a`).

- **Curvature of the Field:** Even if a lens is corrected for spherical aberration, coma, and astigmatism, the image of a flat, extended plane object perpendicular to the optic axis is not a flat plane, but a **curved paraboloid surface** concave toward the lens. This occurs because off-axis object points are farther from the optical center of the lens than axial points, so their images focus closer to the lens (since marginal focal length is shorter than paraxial focal length).
- **Minimization (Petzval's Condition):**
  1. For a single lens, curvature is reduced by placing an aperture stop at a suitable position in front of the lens.
  2. For a system of thin lenses of focal lengths $f_1, f_2, \ldots, f_n$ and refractive indices $\mu_1, \mu_2, \ldots, \mu_n$, the radius of curvature R of the final image surface is given by $\frac{1}{R} = \sum \frac{1}{\mu_n f_n}$. For the image field to be completely flat ($R = \infty \implies \frac{1}{R} = 0$), the lenses must satisfy **Petzval's condition**:

$$\boxed{\frac{1}{\mu_1 f_1} + \frac{1}{\mu_2 f_2} = 0 \implies \mu_1 f_1 + \mu_2 f_2 = 0}$$

  Since $\mu_1, \mu_2 > 0$, f_1 and f_2 must have opposite signs (one convex lens and one concave lens).

1.3 Chromatic Aberration: Longitudinal, Transverse & Circle of Least Chromatic Aberration

Exam Appearances: `2011-12 Q4a (15)`, `2012-13 Q6c (10)`, `2016-17 Q4a (10)`, `2017-18 Q4a (10)`, `2023-24 Q5a (10)`.
Textbook Citations: *Physics for Engineers – Part 1*, Dr. Giasuddin Ahmad, Ch. "Chromatic Aberration"; Dr. Md. Azizar Rahman, BUET PHY 121 Lecture Slides (*All Lecture of Defect of images.pdf*, Slides 54–64).

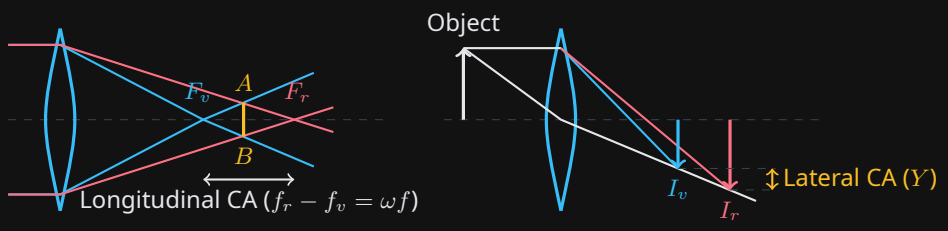
1. Define Chromatic Aberration, Longitudinal and Transverse Chromatic Aberration, and Circle of Least Chromatic Aberration.

- **Chromatic Aberration:** When a beam of white light passes through a single lens, the lens fails to focus all constituent colors to a single point because the refractive index of glass varies with wavelength ($\mu_v > \mu_r \implies f_v < f_r$ by the lensmaker's equation $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$). Consequently, the violet image forms closest to the lens and the red image forms farthest from the lens, producing colored fringes around the image.
- **Longitudinal (Axial) Chromatic Aberration:** The axial distance along the principal axis between the violet focus and the red focus:

  $\boxed{x = f_r - f_v = \omega f}$ (for a parallel beam), or $\boxed{v_r - v_v = \dfrac{\omega v^2}{f}}$ (for a finite object distance u)

  where $\omega = \frac{\mu_v - \mu_r}{\mu - 1}$ is the dispersive power of the lens glass and $f = \sqrt{f_v f_r}$ is the mean focal length.
- **Transverse (Lateral) Chromatic Aberration:** Because the focal length differs for different wavelengths, the lateral magnification $m = \frac{v}{u}$ also varies with color. The difference in vertical height $Y = y_r - y_b$ between the red and blue/violet images of an off-axis object is called **transverse (or lateral) chromatic aberration**.
- **Circle of Least Chromatic Aberration:** Between the violet focus P and the red focus Q along the principal axis, there is a plane XY where the diverging violet cone intersects the converging red cone, forming a circular patch of minimum diameter $AB = d$ where chromatic blurring is least noticeable. For a lens of aperture diameter D, the diameter of this circle is $\boxed{d = \dfrac{1}{2} D\omega \dfrac{v}{f}}$ (which equals $\frac{1}{2}D\omega$ for parallel incident rays where $v = f$).

(a) Longitudinal CA & Circle of Least CA **(b) Transverse (Lateral) CA**



2. How can Chromatic Aberration be Minimized (Achromatism)?

1. **Achromatic Doublet (Two Thin Lenses in Contact):** Cementing a converging crown-glass lens ($+f_1, \omega_1$) in contact with a diverging flint-glass lens ($-f_2, \omega_2$) such that $\boxed{\dfrac{\omega_1}{f_1} + \dfrac{\omega_2}{f_2} = 0}$ eliminates chromatic aberration (commonly used in telescope objectives).
2. **Two Thin Lenses of the Same Glass Separated by a Distance d:** Placing two convex lenses made of the same glass ($\omega_1 = \omega_2$) at a separation equal to the mean of their focal lengths ($\boxed{d = \dfrac{f_1 + f_2}{2}}$) eliminates chromatic aberration (commonly used in Huygens and Ramsden eyepieces).

2 Interference of Light

2.1 Conditions for Sustained Interference, Energy Conservation & Coherent Sources

Exam Appearances: `2009-10 Q5a,c (11), 2011-12 Q5a (8), 2012-13 Q4a (7), 2013-14 Q5a (4), Jan 2020 Q1a (16), 2021-22 Q5a (10), 2023-24 Q6a (10), 2024-25 Q2a (8).`
Textbook Citations: *Physics for Engineers – Part 1*, Dr. Giasuddin Ahmad, Ch. "Interference of Light"; Dr. Md. Azizar Rahman, BUET PHY 121 Lecture Slides (*All Lecture of Interference.pdf*, Slides 4–6, 15–17, 31, 35–36).

1. What is Interference of Light? State the Conditions for Sustained Interference.

- **Definition:** The modification in the distribution of light intensity (producing alternate bright and dark fringes) in the region of superposition of two or more light waves from coherent sources is called **interference of light**.
- **Conditions for Sustained (Well-Defined, Stationary) Interference:**
  1. The two interfering light sources must be **coherent** (they must maintain a constant phase relationship over time, which requires deriving both beams from a single parent source).
  2. The two waves must have the **same frequency and wavelength** (monochromatic light).
  3. The amplitudes of the two interfering waves should be **equal** (or nearly equal) so that the minima are completely dark ($I_{\text{min}} = (a_1 - a_2)^2 = 0$), providing maximum fringe contrast.
  4. The two sources must be **narrow** and placed **very close to each other** (small d) while the screen is placed at a large distance ($D \gg d$) so that the fringe width $\beta = \frac{\lambda D}{d}$ is large enough to be clearly resolved.
  5. If the interfering waves are polarized, they must be in the **same state of polarization**.

2. What Happens to the Energy When Two Light Waves Interfere Destructively? Show That Conservation of Energy is Obeyed.

When two coherent waves of intensities $I_1 = a_1^2$ and $I_2 = a_2^2$ superpose with phase difference ϕ, the resultant intensity is:

$$I(\phi) = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$$

- At points of **destructive interference** ($\cos\phi = -1$), the intensity falls to $I_{\text{min}} = (\sqrt{I_1} - \sqrt{I_2})^2$ (which is 0 if $I_1 = I_2 = I_0$). The missing energy $2\sqrt{I_1 I_2}$ is **not destroyed**; instead, it is redistributed to the points of **constructive interference** ($\cos\phi = +1$), where the intensity rises to $I_{\text{max}} = I_1 + I_2 + 2\sqrt{I_1 I_2}$ (which is $4I_0$ when $I_1 = I_2 = I_0$).
- Averaging the resultant intensity over a complete fringe cycle ($0 \le \phi \le 2\pi$):

$$I_{\text{avg}} = \frac{1}{2\pi}\int_0^{2\pi}\left(I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi\right)d\phi = \boxed{I_1 + I_2} \quad (\text{or } 2I_0 \text{ when } I_1 = I_2 = I_0)$$

  Since the average intensity across the screen is exactly equal to the sum of the individual intensities ($I_1 + I_2$) that would be present without interference, **energy is strictly conserved**—interference merely redistributes light energy from dark fringes to bright fringes.

3. How Are Coherent Sources Realized in Thin Films, Newton's Rings, Fresnel's Biprism, and Lloyd's Mirror?

Two independent light sources can never be coherent because atomic light emission occurs in random bursts lasting $\sim 10^{-8}$ s. Coherent sources are realized by splitting a single parent wave via two methods:

1. **Division of Wavefront (Fresnel's Biprism, Young's Double Slit, Lloyd's Mirror):**
   - *Fresnel's Biprism (2021-22 Q5a):* Consists of two acute-angled prisms of very small refracting angle ($\alpha \approx 30'$) joined base-to-base (obtuse angle $\approx 179°$). Light from a single narrow vertical slit S falls symmetrically on both halves; refraction bends the upper and lower halves of the wavefront toward the axis, producing two virtual coherent sources S_1 and S_2 separated by $d = 2a(\mu - 1)\alpha$.
   - *Lloyd's Single Mirror vs. Young's Double Slit (Jan 2020 Q1a):* In Lloyd's mirror, a single slit S_1 and its virtual image S_2 formed by grazing reflection from a polished plane mirror act as coherent sources. **Similarity:** Both use division of wavefront to produce equally spaced straight fringes of width $\beta = \frac{\lambda D}{d}$. **Dissimilarity:** In Young's double slit, both beams travel through air with no reflection phase shift, so the central fringe ($\Delta = 0$) is **bright**. In Lloyd's mirror, the reflected beam suffers a phase reversal of π (extra path difference $\frac{\lambda}{2}$) upon reflection from the optically denser mirror surface, so the central contact fringe is **dark**.
2. **Division of Amplitude (Thin Films & Newton's Rings, 2023-24 Q6a, 2024-25 Q2a):**
   - When a ray from an extended monochromatic source strikes a thin transparent film (or the wedge-shaped air film between the plano-convex lens and flat glass plate in Newton's rings), partial reflection and partial refraction at the upper and lower surfaces divide the amplitude of the incident wave into two reflected rays derived from the same incident ray, making them mutually coherent.
   - *Why the Center of Newton's Rings is Dark:* At the central point of contact, the air-film thickness is $t = 0$, but the ray reflected from the lower air-to-glass (rarer-to-denser) boundary suffers a π phase change (equivalent to an extra optical path difference of $\frac{\lambda}{2}$), causing destructive interference at the center.

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3 Diffraction of Light

3.1 Diffraction Conditions, Fresnel vs. Fraunhofer, Grating & Resolving Power

Exam Appearances: 2009-10 Q4a (10), 2011-12 Q6a,b (18), 2012-13 Q5a (15), 2013-14 Q4a (3), 2014-15 Q8a (part), 2015-16 Q5a (10), 2016-17 Q5a,b,c (22), 2017-18 Q5a,b (15), 2018-19 Q5a,b (15), Jan 2020 Q2a, 2020-21 Q5a (10), 2021-22 Q4a (8), 2024-25 Q3a (8).
Textbook Citations: *Physics for Engineers – Part 1*, Dr. Giasuddin Ahmad, Ch. "Diffraction of Light"; Dr. Md. Azizar Rahman, BUET PHY 121 Lecture Slides (*All Lecture of Diffraction.pdf*, Slides 3–8, 40, 46–49).

1. What is Diffraction of Light? State the Necessary Condition and Effect of Wavelength.

- **Definition:** The bending or spreading out of light waves around the sharp edges of an obstacle or aperture and their encroachment into the region of the geometrical shadow is called **diffraction of light**.

- **Necessary Condition for Diffraction (`2024-25 Q3a`):** The size of the obstacle or width of the slit a must be comparable to the wavelength λ of the incident light ($a \approx \lambda$). If $a \gg \lambda$, rectilinear propagation dominates and bending is negligible.
- **Effect of Replacing Yellow Light with Smaller Wavelength (`2011-12 Q6a`):** Since the angular half-width of the central maximum ($\sin\theta_1 = \frac{\lambda}{a}$) and fringe spacing ($\beta = \frac{f\lambda}{a}$) are directly proportional to λ, replacing yellow light with a much smaller wavelength causes the diffraction fringes to **shrink and crowd closer together**, narrowing the central maximum.
- **Daily-Life Examples/Applications (`Jan 2020 Q2a`):** Iridescent rainbow colors on soap bubbles and thin oil films (interference), rainbow reflection from CD/DVD tracks and butterfly wings (diffraction grating), anti-reflective coatings on camera lenses, and hologram security strips.

2. Distinguish Between Fresnel and Fraunhofer Classes of Diffraction.

Feature	Fresnel Diffraction	Fraunhofer Diffraction
1. Source & Screen Distance	Source and screen are at **finite distances** from the diffracting aperture/obstacle.	Source and screen are effectively at **infinite distances** from the aperture.
2. Wavefront Type	Incident and diffracted wavefronts are **spherical or cylindrical**.	Incident and diffracted wavefronts are **plane wavefronts**.
3. Use of Lenses	**No convex lenses** are used to render rays parallel or focused.	**Two converging lenses** are used (one to collimate light, one to focus onto the screen).
4. Mathematical Treatment	Requires Fresnel half-period zones / Cornu's spiral (complex).	Simple integration of plane wavelets with linear phase variation.

3. What is a Diffraction Grating? Write the Grating Equation and Absent Spectra Condition.

- **Diffraction Grating:** An optically flat glass plate on which a very large number of equidistant, extremely narrow parallel slits of equal width a separated by equal opaque spaces of width b are ruled with a diamond point (typically $15{,}000$ lines/inch).
- **Grating Equation:**

$$\boxed{(a+b)\sin\theta = n\lambda}$$

  where a is the width of each transparent slit, b is the width of each opaque ruling, $d = a+b$ is the **grating element** (reciprocal of the number of lines per unit length, $d = \frac{1}{N'}$), θ is the angle of diffraction for the n-th principal maximum, $n = 0, \pm1, \pm2, \ldots$ is the order of the spectrum, and λ is the wavelength of light.
- **Absent Spectra when $a = b$ (`2014-15 Q8a`):** Missing orders occur when $(a+b)\sin\theta = n\lambda$ and $a\sin\theta = m\lambda$ are satisfied simultaneously, giving $\frac{a+b}{a} = \frac{n}{m}$. If the transparent and opaque widths are equal ($a = b$), then $\frac{a+b}{a} = 2 \implies \boxed{n = 2m}$. Therefore, all **even-order spectra** ($n = 2, 4, 6, \ldots$) will be absent.
- **Resolving Power of an Optical Instrument & Grating:** The ability of an optical instrument to form distinctly separate images of two very close point objects (or of a grating to separate two close spectral lines λ and $\lambda + d\lambda$, defined as $R = \frac{\lambda}{d\lambda} = nN$). By **Rayleigh's Criterion**, two images are just resolved when the principal maximum of one diffraction pattern falls exactly on the first minimum of the other.

4 Polarization of Light

4.1 Polarized vs. Unpolarized Light, Brewster's Law, O/E Rays & Optical Activity

Exam Appearances: 2009-10 Q6a,b (10), 2012-13 Q6a (10), 2013-14 Q6a (2.5), 2014-15 Q8c,d (11), 2015-16 Q6a (12), 2016-17 Q6a,c (20), 2017-18 Q6a,c (18), 2018-19 Q6a,b (25), 2020-21 Q6a,c (20), 2023-24 Q4a (10).
Textbook Citations: *Physics for Engineers – Part 1*, Dr. Giasuddin Ahmad, Ch. "Polarization of Light"; F. A. Jenkins & H. E. White, *Fundamentals of Optics*, Ch. 24, 26, 28.

1. Distinguish Between Unpolarized and Plane-Polarized Light. Why Can Sound Waves Not Be Polarized?

- **Unpolarized (Ordinary) Light:** A transverse electromagnetic wave in which the electric field vector $\vec{E}$ vibrates randomly in **all possible directions** perpendicular to the direction of wave propagation, showing complete axial symmetry.
- **Plane-Polarized (Linearly Polarized) Light:** A transverse light wave in which the vibrations of the electric field vector $\vec{E}$ are restricted to a **single fixed plane** containing the direction of propagation (asymmetric about the ray axis).
- **Why Sound Waves Cannot Be Polarized (**2017-18 Q6a**,** 2018-19 Q6a**,** 2020-21 Q6a**):** Polarization is an exclusive property of **transverse waves** (such as light), where vibrations are perpendicular to the direction of travel and can be restricted to a specific transverse plane. Sound waves in air/fluids are **longitudinal mechanical waves** in which particles oscillate back and forth *parallel* to the direction of wave propagation. Since the oscillations are already along the axis of propagation in all transverse orientations, they are symmetrical about the direction of propagation and cannot be polarized.
- **Processes to Polarize Unpolarized Light (**2013-14 Q6a**):** (1) Polarization by reflection (Brewster's law), (2) Polarization by refraction (pile of plates), (3) Polarization by double refraction (birefringence in calcite / Nicol prism), (4) Polarization by selective absorption / dichroism (tourmaline crystal / Polaroid sheets), and (5) Polarization by scattering.

2. Polarization by Reflection & Brewster's Law (2015-16 Q6a**,** 2016-17 Q6a**,** 2018-19 Q6b**)**

When unpolarized light is incident on the surface of a transparent dielectric medium (such as glass of refractive index μ) at a specific angle of incidence θ_p called the **polarizing angle** (or Brewster's angle), the reflected beam is **completely plane-polarized** with its electric vector vibrating perpendicular to the plane of incidence, while the refracted beam is partially polarized.

- **Brewster's Law Statement:** The tangent of the polarizing angle θ_p is numerically equal to the refractive index μ of the reflecting medium:

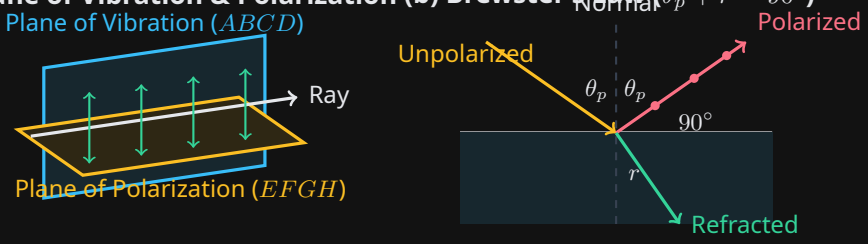
$$\boxed{\mu = \tan\theta_p}$$

- **Proof that Reflected and Refracted Rays Are Perpendicular:** By Snell's law, $\mu = \frac{\sin\theta_p}{\sin r}$. Equating this with Brewster's law $\mu = \tan\theta_p = \frac{\sin\theta_p}{\cos\theta_p}$ gives $\cos\theta_p = \sin r = \cos(90^\circ - r) \implies \boxed{\theta_p + r = 90^\circ}$, so the reflected and refracted rays are mutually perpendicular.

3. Define Optic Axis, Plane of Vibration, Plane of Polarization, and Distinguish Between O-Ray and E-Ray (2009–10 Q6b, 2014–15 Q8c, d).

- **Optic Axis:** The direction of symmetry inside a doubly refracting crystal (passing through the blunt corner of calcite making equal angles with all three faces) along which the O-ray and E-ray travel with the **same velocity** and no double refraction occurs.
- **Plane of Vibration:** The plane containing the direction of propagation and the direction of electric field vibrations ($\vec{E}$) of plane-polarized light (e.g., plane $ABCD$ in the diagram below).
- **Plane of Polarization:** The plane containing the direction of propagation that is strictly **perpendicular** to the plane of vibration, in which no electric vibrations occur (e.g., plane $EFGH$).

(a) Plane of Vibration & Polarization (b) Brewster's Law ($\theta_p + r = 90°$)



Ordinary Ray (O-Ray)	Extraordinary Ray (E-Ray)
1. Obeys the ordinary laws of refraction (Snell's law $\frac{\sin i}{\sin r} = \mu_O =$ constant).	1. Does **not** obey Snell's law in general ($\frac{\sin i}{\sin r}$ varies with direction).
2. Travels with the **same speed in all directions** inside the crystal (spherical wavefront).	2. Travels with **different speeds in different directions** (ellipsoidal wavefront).
3. Plane-polarized with vibrations **perpendicular** to the principal section of the crystal.	3. Plane-polarized with vibrations **parallel** to (in the plane of) the principal section.

4. Explain Optical Activity, Specific Rotation, Its Factors & Applications (2016–17 Q6c, 2017–18 Q6c, 2020–21 Q6c, 2023–24 Q4a).

- **Optical Activity:** The property of certain crystals (e.g., quartz) and solutions (e.g., cane sugar, turpentine, tartaric acid) to rotate the plane of polarization (and plane of vibration) of plane-polarized light passing through them about the direction of propagation is called **optical activity**. Substances that rotate the plane clockwise (looking toward the source) are **dextrorotatory (d)**, and those that rotate it counter-clockwise are **levorotatory (l)**.
- **Specific Rotation (S):** Defined as the angle of rotation θ (in degrees) produced by a 1 decimeter (10 cm) column of solution containing 1 g of optically active solute per cm^3 of solution at a given temperature and wavelength:

$$\boxed{S = \frac{\theta}{l \cdot c}} \quad \left(\text{unit: deg} \cdot \text{dm}^{-1} \cdot \text{g}^{-1} \cdot \text{cm}^3\right)$$

- **Factors on Which Optical Rotation Depends:** (1) Thickness/length of the medium l ($\theta \propto l$), (2) Concentration of the solution c ($\theta \propto c$), (3) Wavelength of light used ($\theta \propto \frac{1}{\lambda^2}$ approximately, Biot's inverse-square law), (4) Temperature of the solution T, and (5) Chemical nature of the substance.

- **Applications:** Saccharimetry in sugar industries to measure sugar purity/concentration, pharmaceutical quality control (distinguishing chiral enantiomers), and medical urinalysis (measuring glucose levels in diabetic patients).

Part II

B) Thermal Physics: Short & Conceptual Questions

5 Thermodynamic Systems, Processes & Kinetic Theory

5.1 Thermodynamic Systems, Processes, Specific Heats & First Law

Exam Appearances: `2009-10 Q1a,b (15), 2013-14 Q1a (10), 2018-19 Q2a (part), 2022-23 Q4a, Q6a (16), 2023-24 Q7a (5).`
Textbook Citations: *Physics for Engineers – Part 1*, Dr. Giasuddin Ahmad, Ch. "First Law of Thermodynamics"; Uploaded Lecture Notes (*2nd Law-Carnot Cycle efficiency.pdf*, Pages 1–3; *Entropy-Reversible and Irreversible Process.pdf*, Pages 6–7).

1. Discuss Different Types of Thermodynamic Systems (`2022-23 Q6a`).

A **thermodynamic system** is a definite quantity of matter bounded by a closed surface (real or imaginary) separating it from its surroundings:

1. **Open System:** Can exchange **both energy (heat and work) and matter (mass)** with its surroundings across its boundary (e.g., a steam turbine, a nozzle, or boiling water in an open vessel).
2. **Closed System:** Can exchange **energy (heat and work) but not matter (mass)** with its surroundings (e.g., a fixed mass of gas enclosed in a cylinder with a movable conducting piston).
3. **Isolated System:** Can exchange **neither energy nor matter** with its surroundings ($\Delta Q = 0, \Delta W = 0, \Delta m = 0$; e.g., a gas enclosed in a rigid, thermally insulated thermos flask, or the universe as a whole).

2. Explain Isothermal, Adiabatic, Isochoric, and Isobaric Processes (`2009-10 Q1a`, `2013-14 Q1a`, `2022-23 Q4a`).

1. **Isothermal Process ($dT = 0$):** A process in which the temperature T of the system remains strictly constant throughout. The gas is enclosed in a cylinder with perfectly conducting walls and compressed or expanded very slowly so heat can freely exchange with the surroundings. For an ideal gas, $PV =$ constant (Boyle's law), $\Delta U = 0$, and $dQ = dW = RT \ln\left(\frac{V_2}{V_1}\right)$.
2. **Adiabatic Process ($dQ = 0$):** A process in which no heat enters or leaves the system ($\Delta Q = 0$). The system is thermally insulated (or the process occurs very rapidly, such as a bursting tire or sound wave propagation). For a reversible adiabatic process of an ideal gas, $PV^\gamma =$ constant and $TV^{\gamma-1} =$ constant, and work is done entirely at the expense of internal energy ($dW = -dU = \frac{P_1V_1 - P_2V_2}{\gamma - 1}$).
3. **Isochoric (Isometric) Process ($dV = 0$):** A process taking place at constant volume (e.g., heating a gas in a rigid sealed container). Since $dV = 0$, no external work is done ($dW = P\,dV = 0$), so all heat supplied increases internal energy ($dQ_V = dU = nC_v dT$) and $\frac{P}{T} =$ constant.

4. **Isobaric Process ($dP = 0$):** A process taking place at constant pressure (e.g., heating a gas in a cylinder with a freely weighted piston or boiling water at atmospheric pressure). Here $\frac{V}{T}$ = constant, work done is $W = P(V_2 - V_1) = nR(T_2 - T_1)$, and $dQ_P = nC_p dT$.

3. Define Internal Energy, C_v, C_p, and State the First Law of Thermodynamics (`2009-10 Q1b`, `2023-24 Q7a`).

- **First Law of Thermodynamics:** When a quantity of heat dQ is supplied to a thermodynamic system Capable of doing external work, the heat absorbed is equal to the sum of the increase in internal energy dU of the system and the external work dW done by the system: $\boxed{dQ = dU + dW = dU + P\,dV}$. (It is the law of conservation of energy applied to thermal processes).
- **Internal Energy (U):** The sum of the intrinsic translational, rotational, and vibrational kinetic energies of the molecules and the intermolecular potential energy of a system. It is a state function (for an ideal gas, U depends only on temperature T).
- **Molar Specific Heat at Constant Volume (C_v):** The amount of heat required to raise the temperature of 1 mole of a gas through 1 K (or 1°C) while keeping its volume constant: $C_v = \left(\frac{\delta Q}{\partial T}\right)_V = \left(\frac{\partial U}{\partial T}\right)_V$.
- **Molar Specific Heat at Constant Pressure (C_p):** The amount of heat required to raise the temperature of 1 mole of a gas through 1 K while keeping its pressure constant: $C_p = \left(\frac{\delta Q}{\partial T}\right)_P = \left(\frac{\partial H}{\partial T}\right)_P$.

5.2 Kinetic Theory Postulates, Brownian Motion, Degrees of Freedom & Maxwell Curves

Exam Appearances: `2011-12 Q1a (8)`, `2012-13 Q1a (8)`, `2018-19 Q1a (12)`, `Jan 2020 Q5a (10)`, `2022-23 Q5a (8)`, `2024-25 Q4a (12)`.
Textbook Citations: *Physics for Engineers – Part 1*, Dr. Giasuddin Ahmad, Ch. "Kinetic Theory of Gases"; N. Subrahmanyam & Brij Lal, *Heat and Thermodynamics* (Uploaded in *Thermometry 2.pdf*, Pages 179–195).

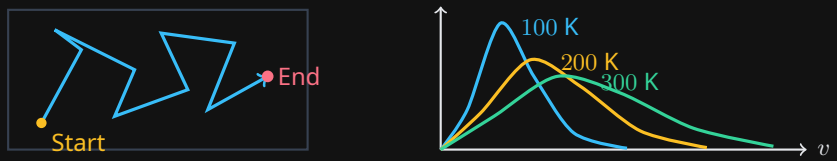
1. State the Fundamental Assumptions (Postulates) of the Kinetic Theory of Gases (`2011-12 Q1a`, `2022-23 Q5a`).

1. Every gas consists of a very large number of identical, tiny, rigid, perfectly elastic spheres called molecules.
2. The molecules are in a state of continuous, rapid, random motion in all directions with velocities ranging from zero to infinity.
3. The actual volume of the molecules is negligible compared to the total volume occupied by the gas (point-mass assumption).
4. No intermolecular forces of attraction or repulsion act between the molecules except during instantaneous collisions.
5. Between two successive collisions, a molecule travels in a straight line with uniform speed; the average distance between collisions is the **mean free path** (λ).
6. Molecular collisions with each other and with the container walls are **perfectly elastic** (both kinetic energy and momentum are conserved), and the duration of a collision is negligible compared to the time between collisions.

2. Define Brownian Motion, Represent It in a Diagram, and State Its Characteristics (2024–25 Q4a).

- **Definition:** The continuous, erratic, zig-zag, random motion executed by microscopic suspended particles (such as pollen grains, smoke particles, or colloidal particles) in a fluid (liquid or gas) due to unequal, random bombardment by the surrounding molecules of the fluid is called **Brownian motion** (discovered by Robert Brown in 1827).
- **Characteristics of Brownian Motion:**
  1. It is completely **continuous, eternal, and random**; no two particles execute the same trajectory at the same instant.
  2. It is independent of external shaking, convection currents, or illumination.
  3. **Effect of Particle Size:** Smaller suspended particles move more rapidly because statistical imbalances in molecular impacts are larger on smaller surface areas.
  4. **Effect of Temperature & Viscosity:** The motion becomes more vigorous at higher temperatures T and in fluids of lower viscosity η (by Einstein's relation, mean squared displacement $\overline{x^2} = \frac{RT}{3\pi N_A \eta r} t$).
  5. It provides direct, visible experimental proof of the kinetic theory of matter.

(a) Brownian Motion Trajectory **(b) Maxwell Curves vs. Temperature T**



3. Degrees of Freedom, Equipartition of Energy, and Maxwell Curve Features (2012–13 Q1a, 2018–19 Q1a, Jan 2020 Q5a)

- **Degrees of Freedom (f or n):** The total number of independent coordinates required to specify completely the position and configuration (or the independent modes of storing kinetic/potential energy) of a dynamical system. For a polyatomic molecule containing N atoms ($3N$ total coordinates):
  - *Linear Polyatomic Molecule:* 3 translational $+2$ rotational $=$ **5** rigid degrees of freedom (plus $3N-5$ vibrational modes, each contributing 2 degrees of freedom at high temperatures).
  - *Non-Linear Polyatomic Molecule:* 3 translational $+3$ rotational $=$ **6** rigid degrees of freedom (plus $3N-6$ vibrational modes at high temperatures).
- **Law of Equipartition of Energy:** For any dynamical system in thermal equilibrium at absolute temperature T, the total energy is distributed equally among all degrees of freedom, and the average energy associated with each molecule per degree of freedom is $\boxed{\frac{1}{2}k_BT}$ (where $k_B = 1.38 \times 10^{-23}$ J/K is Boltzmann's constant).
- **Meaning of Peak, Area, and Temperature Shift in Maxwell's Distribution Curves (Jan 2020 Q5a):**
  1. *Peak of the Curve:* The abscissa (speed v) corresponding to the highest peak of the curve gives the **most probable speed** $v_p = \sqrt{\frac{2k_BT}{m}}$, possessed by the largest fraction of gas molecules.

2. *Area Under the Curve:* Since $f(v)dv = \frac{dN}{N}$ represents the probability of finding a molecule with speed between v and $v + dv$, the total area under each curve from $v = 0$ to ∞ is constant and equal to **unity (**1**)** (or N if dN/dv is plotted).
3. *Why the Curve Flattens and Shifts Right with Temperature:* As T increases, $v_p \propto \sqrt{T}$ shifts to higher speeds (to the right). Because the total area under the curve must remain constant ($= 1$), widening the speed distribution to the right requires the peak height ($f_{\text{max}} \propto T^{-1/2}$) to drop, making the curve broader and flatter at higher temperatures.

5.3 Van der Waals Corrections, Reversible/Irreversible Processes, 2nd Law & Entropy

Exam Appearances: `2009-10 Q2a, Q3a (18), 2011-12 Q2a, Q3a (14), 2012-13 Q2a, Q3a (14), 2013-14 Q2a, Q3a (13), 2014-15 Q5b (12), 2015-16 Q7a, Q8a,b (27), Jan 2020 Q4a (8), 2020-21 Q8a (10), 2021-22 Q8c (7), 2023-24 Q8a (10), 2024-25 Q5a, Q6a (15).`
Textbook Citations: *Physics for Engineers – Part 1*, Dr. Giasuddin Ahmad, Ch. "Real Gases" & "Second Law of Thermodynamics"; Uploaded Notes (*2nd Law-Carnot Cycle efficiency.pdf*, Pages 4–14; *Entropy-Reversible and Irreversible Process.pdf*, Pages 1–4; *Thermometry 2.pdf*, Pages 194–197).

1. Main Considerations Behind the Corrections to the Ideal Gas Equation (`2011-12 Q2a, 2013-14 Q2a`)

Two assumptions of the kinetic theory of ideal gases fail at high pressures and low temperatures:

1. **Finite Size of Molecules (Volume Correction b):** Gas molecules are not mathematical points; they have a finite diameter d and an excluded sphere of influence of volume $4\times$ their actual molecular volume (b). Thus, the compressible free space available for molecular motion is $V - b$ rather than V.
2. **Intermolecular Attractive Forces (Pressure Correction $\frac{a}{V^2}$):** At high pressures, molecules are close together and exert cohesive van der Waals attractions on each other. A molecule about to strike the wall is pulled inward by bulk molecules, reducing its momentum impact. The observed pressure P is less than the ideal pressure by $\Delta P \propto \rho^2 \propto \frac{1}{V^2} = \frac{a}{V^2}$, giving $P_{\text{ideal}} = P + \frac{a}{V^2}$.

2. Reversible vs. Irreversible Processes & Second Law of Thermodynamics

- **Reversible Process:** A quasi-static, non-dissipative thermodynamic process that can be reversed in direction by an infinitesimal change in external conditions so that both the system and its surroundings return to their exact initial states without leaving any permanent change elsewhere in the universe (e.g., slow isothermal or adiabatic expansion in a frictionless cylinder, phase change at constant T and P).
- **Irreversible Process:** Any natural, spontaneous, or rapid process that cannot be retraced in the reverse order without leaving permanent changes in the surroundings, accompanied by an increase in universe entropy ($\Delta S_{\text{univ}} > 0$; e.g., free expansion of a gas into vacuum, Joule heating I^2Rt, friction, diffusion, heat conduction across a finite ΔT).
- **Statements of the Second Law of Thermodynamics:**
  - *Kelvin-Planck Statement:* "It is impossible to construct an engine which, operating in a complete cycle, produces no effect other than the extraction of heat from a single thermal reservoir and the performance of an equivalent amount of work."

- *Clausius Statement:* "It is impossible for a self-acting machine working in a cyclic process, unaided by any external agency, to transfer heat from a body at a lower temperature to a body at a higher temperature."

3. Source vs. Sink Temperature Modification in a Carnot Engine (`2014-15 Q5b`)

The efficiency of a Carnot engine operating between source T_1 and sink T_2 ($T_1 > T_2$) is $\eta = 1 - \frac{T_2}{T_1}$.

- **Method 1: Increasing Source Temperature T_1 by ΔT:**

$$\eta_1 = 1 - \frac{T_2}{T_1 + \Delta T} = \frac{T_1 - T_2 + \Delta T}{T_1 + \Delta T}$$

- **Method 2: Lowering Sink Temperature T_2 by the Same ΔT:**

$$\eta_2 = 1 - \frac{T_2 - \Delta T}{T_1} = \frac{T_1 - T_2 + \Delta T}{T_1}$$

- **Comparison:** Both η_1 and η_2 have the exact same positive numerator $(T_1 - T_2 + \Delta T)$, but the denominator of η_2 is smaller ($T_1 < T_1 + \Delta T$). Therefore:

$$\boxed{\eta_2 > \eta_1} \qquad \left(\text{also confirmed by } \left|\frac{\partial \eta}{\partial T_2}\right| = \frac{1}{T_1} > \left|\frac{\partial \eta}{\partial T_1}\right| = \frac{T_2}{T_1^2}\right)$$

  Hence, **lowering the sink temperature T_2 is more effective and preferred** for increasing the efficiency of a Carnot engine.

4. Representing Any Reversible Cycle as a Cascade of Carnot Cycles (`2021-22 Q8c`)

Any smooth reversible closed cycle ABA on a P-V diagram can be divided into a large number of narrow vertical strips by drawing a series of closely spaced reversible adiabatic curves intersecting the cycle. The top and bottom boundaries of each narrow strip can be approximated by infinitesimal isothermal segments at temperatures T_{1i} and T_{2i}, turning each strip into an elementary Carnot cycle. Along every interior shared adiabatic boundary, adjacent elementary cycles are traversed in opposite directions, so their work and heat contributions cancel out completely—leaving only the outer zig-zag perimeter, which converges to the original reversible cycle as the number of strips approaches infinity ($\sum \frac{\delta Q_i}{T_i} = \oint \frac{dQ_{\text{rev}}}{T} = 0$).

5. Physical Significance of Entropy, Graphical Notation (T-S Diagram) & Clausius-Clapeyron Applications

- **Physical Significance of Entropy & Why Universe Entropy Increases (`2015-16 Q7a`, `2020-21 Q8a`):**
  1. *Measure of Molecular Disorder/Randomness:* By Boltzmann's relation $S = k_B \ln W$, entropy measures the microscopic randomness or disorder of a system.
  2. *Measure of Unavailable Energy:* Entropy measures the degradation of thermal energy into forms unavailable for conversion into mechanical work.
  3. *Why Entropy of the Universe Always Increases:* In an ideal reversible process, $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = \frac{dQ}{T} - \frac{dQ}{T} = 0$. However, all real natural processes (heat transfer across a finite temperature difference $T_1 > T_2$, friction, free expansion) are **irreversible**, so $\Delta S_{\text{univ}} = \frac{Q}{T_2} - \frac{Q}{T_1} = Q\left(\frac{T_1 - T_2}{T_1 T_2}\right) > 0$. Thus, the entropy of the universe continuously increases toward a maximum.

4. *Graphical Notation of Entropy (T-S Diagram, 2015-16 Q7a):* Since $dQ_{\text{rev}} = T\,dS$, plotting absolute temperature T (ordinate) against entropy S (abscissa) gives a **Temperature-Entropy (T-S) diagram** where the area under any reversible curve $\int_{S_1}^{S_2} T\,dS$ represents the heat Q exchanged, an isothermal process is a horizontal line, an adiabatic (isentropic) process is a vertical line, and a Carnot cycle is a simple rectangle of area $(T_1 - T_2)(S_2 - S_1) = W$.

- **Physical Significance of Clausius-Clapeyron Equation ($\frac{dP}{dT} = \frac{L}{T(v_2 - v_1)}$) on Melting, Boiling & Pressure Cookers (2015-16 Q8b, Jan 2020 Q4a):**
    1. *Effect of Pressure on Boiling Point:* During vaporization of any liquid, the vapor volume is much larger than the liquid volume ($v_2 = v_{\text{vap}} \gg v_1 = v_{\text{liq}}$), so $v_2 - v_1 > 0$ and $\frac{dP}{dT} > 0$. Thus, the boiling point of a liquid always **increases with an increase in pressure**.
    2. *Why Food Cooks Faster in a Pressure Cooker (Jan 2020 Q4a):* Inside a sealed pressure cooker, trapped steam raises the internal pressure to ~ 2 atm, which raises the boiling point of water from 100°C to about 120°C (since $\frac{dP}{dT} > 0$). Furthermore, steam has a very high latent heat of vaporization ($L \approx 540$ cal/g $= 2260$ kJ/kg); when steam condenses on the cooler food at this elevated temperature, it rapidly transfers large amounts of thermal energy without drying the food, cooking it much faster.
    3. *Effect of Pressure on Melting Point:* For substances that expand on melting (like wax or sulfur, $v_2 > v_1$), $\frac{dP}{dT} > 0$ so the melting point rises with pressure. For anomalous substances that **contract on melting** (like ice, where $v_{\text{water}} = 1.000$ cm^3/g $< v_{\text{ice}} = 1.091$ cm^3/g, so $v_2 - v_1 < 0$), $\frac{dP}{dT} < 0$, meaning the melting point of ice is **lowered by an increase in pressure** (regelation).

Part III

C) Waves and Oscillations: Short & Conceptual Questions

6 Simple Harmonic Motion, Springs & Lissajous Figures

6.1 SHM Characteristics, Cutting a Spring in Half & Lissajous Figures

Exam Appearances: `2009-10 Q7a (part)`, `2015-16 Q1a,c (15)`, `2016-17 Q1a (7)`, `2020-21 Q1a (8)`, `2021-22 Q1a (8)`, `2022-23 Q1a (8)`, `2023-24 Q1a (7)`.
Textbook Citations: *Physics for Engineers – Part 2*, Dr. Giasuddin Ahmad, Ch. "Simple Harmonic Motion"; Prof. Dr. Md. Rafi Uddin, BUET Lecture Slides (*oldWaves-and-Oscillation-Merged.pdf*, Slides 11–16, 48–51, 78–81); A. P. French, *Vibrations and Waves*.

1. Effect of Cutting a Spring in Half on Oscillation Frequency (`2015-16 Q1c`)

For a uniform spring of natural length L and force constant k, the extension Δx under tension F is proportional to its original length L ($\Delta x \propto L$), so the spring constant is inversely proportional to length ($k \propto \frac{1}{L}$, or $kL =$ constant).

- Before cutting: Original frequency is $f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$.
- After cutting in half ($L' = \frac{L}{2}$): The force constant of each half doubles ($k' = 2k$). Suspending the same mass m from one half gives:

$$\boxed{f' = \frac{1}{2\pi}\sqrt{\frac{k'}{m}} = \frac{1}{2\pi}\sqrt{\frac{2k}{m}} = \sqrt{2}\, f \approx 1.414\, f}$$

  Thus, the frequency is **not the same**; it increases by a factor of $\sqrt{2}$.

2. What Are Lissajous Figures? On What Factors Do They Depend, and What Are Their Applications?

- **Definition:** When a particle is simultaneously acted upon by two mutually perpendicular simple harmonic vibrations, the closed or open trajectory traced by the resultant motion of the particle is called a **Lissajous figure**.
- **Factors on Which Lissajous Figures Depend:**
  1. The **ratio of the frequencies** (or time periods, $\omega_1 : \omega_2$) of the two component vibrations.
  2. The **ratio of the amplitudes** ($a : b$) of the two component vibrations.
  3. The **initial phase difference** (δ or ϕ) between the two component vibrations.
- **Applications of Lissajous Figures:**
  1. *Accurate Determination of an Unknown Frequency:* Applying a known frequency f_x to the horizontal plates and an unknown frequency f_y to the vertical plates of a Cathode Ray Oscilloscope (CRO) gives a stationary loop pattern where $\boxed{\frac{f_x}{f_y} = \frac{n_y}{n_x}}$ (n_y, n_x = number of tangency points along the vertical and horizontal boundaries).

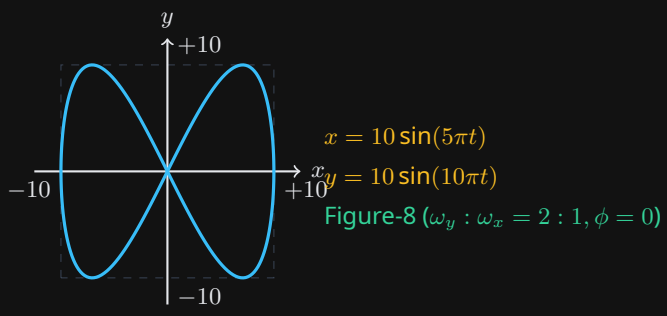
2. *Measurement of Phase Difference:* Measuring the y-intercept and maximum amplitude of the $1:1$ ellipse on a CRO determines the phase shift ϕ introduced by an electrical network or amplifier.
3. *Synchronization of Tuning Forks and Audio Oscillators.*

3. Lissajous Figure for $x = 10\sin(5\pi t)$ and $y = 10\sin(10\pi t)$ (2020-21 Q1a)

Here $\omega_y = 10\pi = 2\omega_x$ (frequency ratio $\omega_y : \omega_x = 2:1$), equal amplitudes $a = b = 10$, and initial phase difference $\phi = 0$. Using $\sin(10\pi t) = 2\sin(5\pi t)\cos(5\pi t)$, we have:

$$\frac{y}{10} = 2\left(\frac{x}{10}\right)\sqrt{1-\left(\frac{x}{10}\right)^2} \implies y^2 = \frac{4x^2}{100}(100 - x^2)$$

This traces a symmetrical **figure-eight (∞)** bounded within $-10 \le x \le 10$ and $-10 \le y \le 10$, touching the horizontal boundary twice ($n_x = 2$) and vertical boundary once ($n_y = 1$).



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7 Damped & Forced Oscillations, Wave Motion & Acoustics

7.1 Free, Damped & Forced Oscillations, Door Closers, Quality Factor & Resonance

Exam Appearances: 2012-13 Q8a (part), 2013-14 Q7a,b (14), 2014-15 Q2a (5), 2015-16 Q2b (5), 2016-17 Q2a (7), 2017-18 Q1a (7), 2018-19 Q7a, Q8a (15), 2020-21 Q2a (8), 2021-22 Q2a (8), 2022-23 Q2a (8), 2023-24 Q2a (5), 2024-25 Q7a, Q8a (20).
Textbook Citations: *Physics for Engineers – Part 2*, Dr. Giasuddin Ahmad, Ch. "Damped and Forced Oscillations"; Prof. Dr. Md. Rafi Uddin, BUET Lecture Slides (*oldWaves-and-Oscillation-Merged.pdf*, Slides 85–113); D. Halliday, R. Resnick & J. Walker, *Fundamentals of Physics*, Ch. 15.

1. Define Free, Damped, and Forced Oscillations. Why Is Free Vibration Impossible in Reality?

1. **Free (Undamped) Oscillations:** Oscillations of a body displaced from equilibrium and allowed to vibrate purely under its own internal elastic restoring force ($F = -kx$) with constant amplitude and natural frequency $\omega_0 = \sqrt{\frac{k}{m}}$, in the complete absence of external resistive or driving forces.
    - *Why Free Vibration is Impossible in Reality (2018-19 Q7a):* Every real physical system experiences dissipative forces such as air drag, fluid viscosity, internal friction at the support, or electromagnetic radiation damping, which continuously convert mechanical energy into thermal energy and cause the amplitude to decay.

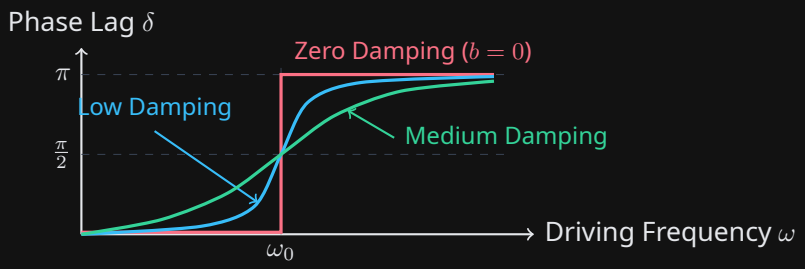
2. **Damped Oscillations:** Oscillations in which the motion is opposed by resistive/frictional forces ($F_d = -bv$), causing the mechanical energy and amplitude ($A(t) = A_0 e^{-(b/2m)t}$) to decrease gradually to zero.
3. **Forced (Driven) Oscillations:** Oscillations in which a body is driven continuously by an external periodic force $F(t) = F_0 \cos(\omega t)$ to compensate for damping losses. After transients die out, the body oscillates at the **frequency of the external driving force (ω)** rather than its own natural frequency (ω_0).

2. Door Closer Damping, Shock Absorbers, Quality Factor (Q), and Phase Lag Graph

- **Which Type of Damping Motion Should Be Set in a Door Closer? (2015–16 Q2b, 2020–21 Q2a): Critical damping ($\gamma = \omega_0$, i.e., $\frac{b}{2m} = \sqrt{\frac{k}{m}}$)** should be set in a door closer. *Justification:* If the door closer is *underdamped* ($\gamma < \omega_0$), the door will slam shut and swing back and forth repeatedly. If it is *overdamped* ($\gamma > \omega_0$), the door will take an inconveniently long time to creep shut. Critical damping returns the door to its closed equilibrium position in the **shortest possible time without slamming or oscillating**.
- **Influence of Damping Coefficient b & Engineering Examples (2024–25 Q7a):** Increasing b causes the amplitude $A(t) = A_0 e^{-(b/2m)t}$ to decay faster and lowers the oscillation frequency $\omega' = \sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2}$. Real-world examples where tuned damping improves safety and performance include **automotive hydraulic shock absorbers** (tuned near critical damping to suppress road bounce immediately) and **seismic tuned mass dampers in bridges and skyscrapers**.
- **Quality Factor (Q) and Power Dissipation (2013–14 Q7a):**
  - *Power Dissipation:* The rate at which mechanical energy is lost to heat against the damping force, $P(t) = -\frac{dE}{dt} = bv^2$, with cycle-averaged value $P_{\text{avg}} = \frac{b}{m}E = 2\gamma E$.
  - *Quality Factor (Q):* A dimensionless measure of the sharpness of resonance and freedom from damping, defined as 2π times the ratio of energy stored to energy dissipated per cycle:

$$\boxed{Q = 2\pi \frac{E}{|\Delta E_{\text{cycle}}|} = \frac{\omega_0}{2\gamma} = \frac{\omega_0 m}{b} = \frac{\sqrt{km}}{b}}$$

- **Resonance, Why Amplitude Peaks at Resonance, and Applications (2018–19 Q8a, 2024–25 Q8a):** When the frequency ω of the external periodic force matches the natural frequency ω_0 of the oscillator ($\omega \approx \omega_0$), the driving force is always in phase with the velocity of the oscillator, transferring maximum positive power into the system on every cycle until energy input balances damping loss at a maximum amplitude $A_{\text{max}} \approx \frac{F_0}{b\omega_0}$. *Applications:* Tuning radio/TV LC receiver circuits to a desired station frequency, Magnetic Resonance Imaging (MRI), musical instruments (acoustic resonance), and structural engineering (avoiding wind/seismic resonance such as the 1940 Tacoma Narrows Bridge collapse).
- **Phase Lag δ of a Driven Oscillator vs. Driving Frequency (2013–14 Q7b):** From $\tan\delta = \frac{P\omega}{\omega_0^2 - \omega^2}$, for $\omega \ll \omega_0$, $\delta \to 0$ (displacement in phase with force); at exact natural frequency $\omega = \omega_0$, $\delta = \frac{\pi}{2}$ ($90°$ lag) for all non-zero damping; and for $\omega \gg \omega_0$, $\delta \to \pi$ ($180°$ out of phase).



7.2 Wave Motion, Progressive vs. Stationary Waves & Architectural Acoustics

Exam Appearances: `2013-14 Q8c (7)`, `2014-15 Q3a (part)`, `2016-17 Q3a (7)`, `2017-18 Q3a (8)`, `Jan 2020 Q8b (10)`, `2021-22 Q3a (8)`, `2022-23 Q3a (8)`, `2023-24 Q3a (7)`.
Textbook Citations: *Physics for Engineers – Part 2*, Dr. Giasuddin Ahmad, Ch. "Wave Motion", "Stationary Waves" & "Acoustics of Buildings"; Prof. Dr. Md. Rafi Uddin, BUET Lecture Slides (*oldWaves-and-Oscillation-Merged.pdf*, Slides 9–10, 124–126).

1. Explain Wave Motion, Types of Waves, and Distinguish Progressive vs. Stationary Waves.

- **Wave Motion:** A form of disturbance that travels through a medium due to the repeated periodic vibrations of the particles of the medium about their mean equilibrium positions, transferring energy and momentum from one point to another without any net transport of matter as a whole.
- **Three Fundamental Types of Waves (`2016-17 Q3a`):**
  1. *Mechanical Waves:* Governed by Newton's laws and require a material medium possessing elasticity and inertia (e.g., sound waves, water waves, waves on a string). Classified into **Transverse waves** (particles vibrate perpendicular to wave propagation, forming crests and troughs) and **Longitudinal waves** (particles vibrate parallel to propagation, forming compressions and rarefactions).
  2. *Electromagnetic Waves:* Consist of coupled oscillating electric and magnetic fields ($\vec{E} \perp \vec{B} \perp \vec{v}$) that require no material medium and travel through vacuum at $c = 3 \times 10^8$ m/s (e.g., light, radio, X-rays).
  3. *Matter Waves (de Broglie Waves):* Quantum-mechanical probability waves associated with moving subatomic particles such as electrons, protons, and neutrons ($\lambda = \frac{h}{mv}$).

Progressive (Traveling) Wave	**Stationary (Standing) Wave**
1. The disturbance travels forward continuously with a definite wave velocity v and transfers energy across every section.	1. The waveform remains confined between fixed boundaries and transfers **zero net energy** across any section.
2. Every particle vibrates with the **same amplitude** a, but phase varies continuously from particle to particle (kx).	2. Amplitude varies with position as $2a\cos(kx)$—zero at **nodes** and maximum ($2a$) at **antinodes**; all particles in a loop vibrate in the same phase.
3. No particle in the medium is permanently at rest.	3. Particles at the nodes are permanently at rest (separated by $\frac{\lambda}{2}$).

2. Solution Check for 1D Wave Equation (`2013-14 Q8c`)

The 1D progressive wave equation is $\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2}$.

- **(i)** $y = 2\sin x \cos(vt) = \sin(x + vt) + \sin(x - vt)$**:** Here $\frac{\partial^2 y}{\partial t^2} = -2v^2 \sin x \cos(vt)$ and $\frac{\partial^2 y}{\partial x^2} = -2\sin x \cos(vt)$, so $\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2}$ is **satisfied** (it is a valid solution).
- **(ii)** $y = \sin(2x)\cos(vt)$**:** Here $\frac{\partial^2 y}{\partial t^2} = -v^2 y$ whereas $v^2 \frac{\partial^2 y}{\partial x^2} = -4v^2 y \neq \frac{\partial^2 y}{\partial t^2}$, so it is **not a solution** for wave speed v.

3. Reverberation, Reverberation Time, and Measures for Good Auditorium Acoustics (`Jan 2020 Q8b`, `2021-22 Q3a`)

- **Reverberation:** The persistence or prolongation of sound in an enclosed room or hall due to multiple reflections from walls, ceiling, floor, and objects even after the source of sound has stopped emitting.
- **Reverberation Time (T):** The time required for the sound intensity (or energy density) in a room to fall to one-millionth (10^{-6}, a 60 dB drop) of its initial steady-state value after the sound source is switched off ($T = \frac{0.161\,V}{\sum a_i S_i}$).
- **Measures for Clear Audibility of Speech or Music in an Auditorium (`Jan 2020 Q8b`):**
  1. **Optimum Reverberation Time:** Maintain $T \approx 0.5$–1.0 s for lecture/speech halls and 1.2–2.0 s for concert/music halls by installing acoustic absorbent materials (heavy curtains, carpets, perforated acoustic tiles, upholstered seats).
  2. **Sufficient Loudness:** Use a parabolic sounding board behind the speaker and a public-address loudspeaker system so sound intensity is adequate throughout the hall.
  3. **Elimination of Echoes and Focusing Defects:** Avoid large concave or domed ceilings/walls that focus sound into "hot spots" and dead zones; cover distant rear walls (path difference > 17 m) with sound absorbers.
  4. **Avoidance of Echelon Effect and Extraneous Noise:** Cover regular stair steps with carpets to prevent musical frequency reflections (echelon effect), and soundproof doors/windows and HVAC ducts against external noise.

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